

ISyE 6501
Introduction to Analytics Modeling
Course Project
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Kt wiz Description

Kt wiz is a South Korean professional baseball team in the KBO League. In the article that I read, their main motive is to keep the stadium filled by using marketing and data visualization. Therefore, the one thing that kt wiz values is fan engagement. They work to keep the fans engaged through the many things they offer at the baseball game. They believe that focusing on fan engagement will help maximize ticket sales and keep the stadium filled.

Potential variables of data / how can it be collected

1. Ticket purchase history: Looking at the past ticket purchase history of the buyer could play a big role in this situation. A person who regularly buys tickets to kt wiz's games could be more likely to come compared to someone's first time.
 - a. Ticket purchase history data can be collected from the ticketing system through the web, mobile app, or POS. The ticket purchase history data may include user ID, ticket ID, purchase date, game data, etc.
2. Seat selection: The location of the seat could affect whether the seat will be bought or not. If the seat is in a bad location, then it is possible that no one will buy that seat for the game.
 - a. Seat selection data can be collected from the app or website. It would show the seat map and seat info.
3. Unattendance: It is good to store data on the number of people who don't show up. This can be viewed as our "error" when comparing the number of tickets bought vs. the number of people who attend (in other words, number of seats filled).
 - a. Attendance frequency data can be collected by percentage. It would be the number of times they show up divided by the total number of times they bought the ticket.
4. Entry times: The time that the fan comes in is also important. To keep the stadium filled, it is important to keep track of whether a seat is vacant or occupied. If there is an empty seat in the middle of the game, there are four possibilities. One: The person was at the game but left early. Two: The person showed up late to the game. Three: The person did not show up at all. Four: The person sat in a different open seat. To classify these, we can perform classification from past data and use an estimate on how many of each there will be in future games with respect to the error.

- a. Entry times data can be collected by setting four different ranges of time: Start of the game to end of the game, start of the game to middle of the game, middle of the game to end of the game, and no show (N/A).
 - b. This information can be collected through timestamped QR scans at the stadium entrances.
5. Concession purchases: Concession purchases is a must when it comes to engaging the fans. No one wants to watch a game with no food or drinks. Having concessions can give fans something to eat or drink while enjoying the game, and it can also make the game a much more enjoyable experience. Not only that, but concessions can help kt wiz gain more profit in sales.
 - a. Concession purchase data can be collected through the mobile app or POS systems.
6. Mobile app usage: kt wiz have mobile app where people can book and pay for their tickets. There are many other features that they can use on the app. There is in-seat food and merchandise ordering, social features, and interactive viewing to zoom in on specific areas of the field and stands.
 - a. The number of times the fan uses the food and drink delivery feature on their app.
 - b. The number of times the fan comes to future games using the promotions
7. Age, gender, location, family status: The demographics of the buyer could also play a big role. There could be patterns in the buyer's age, gender, location, and family status and how this information could affect the ticket sales "indirectly".
 - a. The demographics data can be collected through account registration.
8. Push notification responses: Having push notification responses on notifies the person about team updates, seat information, promotions, and more. If people have this on, then they can see updates on the game much quicker and clearer, and they are more likely to buy promotions and deals since notifications are an effective strategy to engage the fan into spending more money at the game.
 - a. Classifying whether the fan has push notification responses on or off (1 for on, 0 for off)
9. Posts, hashtags, likes, mentions during game: The number of posts, hashtags, likes, and mentions on social media shows how engaged the fan is. Additionally, it is another way of promoting kt wiz's baseball games and inviting other new individuals to come to the games in the future. It could also increase their fan base.
 - a. Social media activity data can be collected by counting the number of each through social listening tools.
10. Ratings, complaints, suggestions: kt wiz can know what they do well and what they need to improve on based on fans' ratings, complaints, and suggestions. Doing so, they can readjust the system of their games and please the fans by reading their comments.
 - a. This data can be collected through feedback forms, surveys, reviews, etc.
11. Weather, opponent team, day of the week, injuries, etc: External factors can come into play. The weather on game day can affect whether people come or not. If it is raining, then people are less likely to come. The opponent team is another factor. If the opponent team is a team that has been performing poorly or don't have well known players, then others are less likely to come. However, if they are against a good team, then people would want to watch that for more entertainment. The day of the week is another possible factor. People would more likely come on weekends rather than weekdays. The only people who would come on weekdays are the youth since older people are most likely at work. Injuries is another possible factor. If a good player on kt wiz is injured, then people may not want to come because they cannot see their

favorite player play on the field. There are many other external factors, but these are only some of them.

- a. Weather data can be collected from weather services every game day.
- b. Opponent team data can be collected by keeping track of which team is playing on game day.
- c. Day of the week data can be collected from game schedules.
- d. Injury data can be collected from injury reports.

What models might be used to create the solution

1. Classifying individuals into whether they show up to the game or not
 - a. Entry times: As I mentioned before, the four possibilities of the entry times of a person are that the person shows up late, leaves late, doesn't show up, or takes another open seat. To classify these individuals, we would use k-means clustering. There would be four different clusters: one for each possibility.
 - b. Unattendance (or attendance frequency): We can also classify individuals based on their attendance frequency. If the individual has a perfect attendance percentage (100%), then they are most likely to show up to the game. If the individual has an attendance percentage of 50%, then they are not likely to show up to the game. We can classify these individuals into two separate groups (1 for showing up, 0 for not showing up). We can also set a threshold for the classification. A good estimator could be that the people who are most likely to show up have an attendance percentage ranging from 80%-100% while the people who are not likely to show up have an attendance percentage lower than 80%.
2. Recommendation system
 - a. To improve game attendance, it's best to reach out to fans and allow as much fan engagement as possible. One of the ways kt wiz utilizes this is through their mobile app. The mobile app allows fans to order their drinks and food on the app and have it delivered to their seat. The mobile app also gives promotions on future mobile ticket purchases or in-app orders so that fans are encouraged to come again next time for better deals. Finally, if the fan has push notification responses on, then they are more likely to be engaged because they get direct, quick notifications to keep them updated. The app does a great job in personalizing content and engaging the fans directly to increase ticket purchases. We could use content-based filtering after the fans are clustered into their groups.
3. Prediction Model
 - a. To determine which fans are likely to stop attending or engaging, we could use the variables: ticket purchase history, unattendance, entry times, concession purchases, mobile app usage, app usage, push notification response, social media mentions, ratings/complaints and external factors such as weather, opponent team, day of the week, etc.

- b. Since we are predicting which fans are likely to stop attending or engaging, we could use a Random Forest model with the classification tree to put fans in two different groups: Likely to attend and engage / Unlikely to attend and engage.
- 4. Attendance Forecasting Model
 - a. We want to predict the stadium attendance for future games so that we know whether the stadium will be filled or not. Since kt wiz's goal is to fill the stadiums up, we can use variables on attendance and fan engagement to predict whether people are most likely to show up to the game or not.
 - b. One analytical model that we could use for this scenario is ARIMA.
 - i. AR: We would use past values of the attendance series to predict future values. For example, if the attendance was high last Saturday, it might be high this Saturday as well. Therefore, attendance at Game t depends on attendance at Game $t - 1$, $t - 2$, etc.
 - ii. I: We could identify the number of differences to make the series stationary. If attendance gradually increases over the season, the differencing will help remove the trend, so the model focuses on the fluctuations.
 - iii. MA: We would use the past forecast residuals to improve the predictions. For example, the prediction residual from Game $t - 1$ helps Game t . This is to adjust underestimations and overestimations.
- 5. Multi-armed Bandit (Optimization)
 - a. The goal is to maximize the success rate of marketing campaigns by finding the best-performing strategies as we explore new ones.
 - b. Campaign Variants
 - i. Push Notification Response
 - ii. Mobile app usage examples
 - 1. Seat upgrade offers
 - 2. Loyalty progress reminder
 - c. Track Reward Metrics
 - i. Open rates (also known as screen time pick-ups)
 - ii. Click-through rates
 - iii. Engagement time
 - d. Multi-armed bandit algorithm
 - i. Upper Confidence Bound
 - e. Analyze results and best performing strategies
- 6. Support Vector Machine
 - a. The purpose is to analyze fan reactions on social media or feedback forms.
 - b. We want to measure the fan sentiment. The variables that we would use are posts, likes, hashtags, mentions, ratings, complaints, and suggestions.
 - i. We would need to convert the text into numeric features and train them on labeled data.
 - c. Since SVM focuses on classification, we can use the numeric features (+1, -1, or 0) to determine which ones are positive and negative words.
 - d. Then, we should tune the C and kernel to see which model gives the best accuracy and precision.

Combining the Models

1. First, we would perform clustering to separate the fans into groups. All cluster suggestions that I mentioned can be used.
2. Personalized recommendations are sent after depending on where those fans are clustered into.
3. Fans' feedback can then be tracked to adjust the models using our support vector machine.
4. We can use our ARIMA model to predict attendance. After prediction, we can run the targeted offers on slow days.
5. We can analyze the response and improve our strategy based on it.
6. Finally, we use our support vector machine model and track the ones that show negative sentiment.

Refreshing Data and Re-running Models

1. For the customer segmentation, we could refresh the data and re-run the models every month. We want to be updated on fans' reviews, ratings, and engagement every month and improve the baseball experience for them.
2. For the recommendation system, we could refresh the data and re-run the models every week. It should adapt to recent fan interactions and preferences with the mobile app.
3. For the prediction model, we could refresh the data and re-run the models every month. Since we need to know when fans are likely to stop attending games, we should refresh it every month to stay up to date.
4. For the ARIMA model, we could refresh the data and re-run the models before every home game. ARIMA is most useful for short term predictions, so it is best to update before every home game and predict the next time kt wiz have a game based on the updated data.
5. For our multi-armed bandit optimization model, we could refresh the data and re-run the models on real time. If our model learns customer engagement through the mobile app overtime, then it can learn more and give faster feedback loop, which results to better targeting.
6. For our sentiment analysis, we could refresh the data and re-run the models on real time. Then, it would track and react to public sentiment immediately. It's best to refresh it as soon as possible. If we want to improve fan experience, improving based on their wants, needs, and criticism is most important.

References

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https://play.google.com/store/apps/details?id=kr.co.ktwiz.wizzap&hl=en_US

